

Rejoinder

Things that changed

- Our opponent said that our presentation of artifacted particles described them more than what we have done in our thesis, so we deleted the text for the artifacted particles and rewrote the text to mimic what we said in our presentation.
- The examiner said in our presentation that she thought that it was unclear why we had a “level of education” question, in our questionnaire. So we tried to make it easier to understand why in the subsection.
- Our opponent said that the word “billboarded” is spelled like so, and not as “billboard-ed”, so the change was made for every instances of that word.
- Our opponent wanted a cleaner specification of when we are explaining computer parts used, and remove “Asus”, so the change was made in the text to display a bit less of every computer part and display a bit cleaner.
- Our opponent said that FPS and OBB should not be abbreviations in the Abstract chapter, so the change was made for every instance of that word to not be shortened in the abstract chapter.
- Our opponent said that skeletal animation is explained wrongly in the abstract chapter, so the change was made to try explaining it correctly like in the future chapters.
- Our opponent said that abstract need to be shorten down, preferably to 1-2 sentences per point or half a page, so the change was made to shorten it down and meet the criteria of half a page.
- Our opponent said that the result description in the abstract text is too short and the information given is very superficial and does not match the results, so the change was made to explain it better and correctly.
- Our opponent said that there is a need for section called “Gap” in the thesis, that tells the reader what skeleton animation cannot do, so the change was made to include the “Gap” section.
- Opponent thought we had too few related works therefore we found an extra related work by Epic Games about an Niagara library in Unreal Engine 5. However, no paper where found, only a presentation, and that presentation where cited.
- Our opponent said that the number of participants should be mentioned in the beginning of section 4.2 so the reader knows when reading the graph, so the change was made to add a short sentence and mention the number of total test subjects participating in the survey questionnaire.
- Our opponent said that even though section 3.2.3.2 was well described, but some words are repeated many times and becomes repetitive, and “could benefit from minor restructuring to improve their clarity and make them easier to understand”, so the change was made to improve the text and modify it a bit.
- Opponent said that the research question (probably meaning thesis questions) were not motivated or explained on why the questions were created. Therefore in the thesis section the questions were given an motivation and explanation.
- Our supervisor thought it would be a good idea to mention Nvidia Flex in related work, we had used Nvidia Flex this in our thesis as an acceleration technique but thought it doesn't fit, instead we moved Nvidia Flex to section “other suspended sediments”, because the library can create fluid particles and collision with particles.

- Removed section about loading particle colors and moved the text into the section when billboarded particles are created.
- removed unnecessary use of the word “author”.
- Our opponent said that most of the graphs were inconsistent and needed to be fixed to look better, so the change was made to fix them according to instructions of the opposition and to our taste of what we want to present in the graphs
- A paragraph was added to explain why the survey was done online, what negatives and what positives exist because of the survey being online instead of being in person.
- Our opponent said that the information given in parenthesis should be incorporated into text, so the change was made to incorporate it into the appropriate text section
- We decided to talk about physical memory and why we did not include it in the graphs, so the change was made to include a bit of information on it.
- A question was asked at the presentation where the bottle neck for the program is, we answered the GPU, and that should probably be in the thesis also, so that is written in the discussion section now.
- Our opponent said that picture, figures and graphs should be mentioned and described in the text, so the change was made to include, talk and refer to them in the text
- Our opponent said that the discussion on performance results was lacking and wanted us to go a bit more in depth, so the change was made to try to expand on explaining and discussing the results of performance.
- Our opponent said that it is never mentioned how sensitive data will be handled, so we made the change to include an section on GDPR and data security, where we explained how such data is handled.
- Opponent said that our methods needs to improve readability therefore we change some of the text to make it easier to understand and read.
- Our supervisor recommended using `\ref{}` in overleaf, so we made the change to use `\ref{}` for every section title, subsection, figure, graph, algorithm and picture.
- Our opponent said he thinks it's unclear about the “future in future media”, so we reviewed the sentence regarding the section of “Appeal and Ratio” and realized it was unnecessary and repetitive and the section got removed.
- The three subsections in the implementation section (preprocess, CPU, and GPU) all got a pseudo code to make it easier for reader to know what's going on and how the implementation was done.
- Our opponent said on the presentation that we need to include an answer if the particles could be stuck in the body of the character, so we expanded and gave an answer for those cases that particles could get stuck in the body.
- Changed the keywords by added the words “Velocity” and “Collision” and removing “Oriented bounding box” because at most 2 keywords could be in the title.
- Our opponent gave us a paper that had notes about our thesis, in the notes it said that we could add spaces in the particle count so instead of saying 4359198 it says 4 359 198 so its easier to read. so we did just that

Things that did not change

- Our opponent suggested that it would be a requirement to include a section on “validity and Reliability”. However, we hold the view that assessing the validity and

reliability within our paper is challenging due to the questionnaire participants not being in person and the limitation of the test being done on a single computer. Our thesis primarily focuses on the idea of POBBSA instead of providing concrete empirical evidence. Our supervisor also agreed with our decision.

- Both supervisor and the opponent suggested that we should have less depth in subsections and maybe have the text in a more fluid way. However, if more subsections would be removed and more fluidity in the text was needed. We would not be able to reference certain sections in the text making the text go on tangents in the middle of sentences. making the text not being easily readable either.